

NONLINEAR DIGITAL SIGNAL/IMAGE PROCESSING: OVERVIEW AND PERSPECTIVES

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ABSTRACT

This paper presents an overview of the various nonlinear digital filter classes that exist in the literature. The properties of these classes are presented as well as their applications. The interrelation of the filter classes and the efforts towards unification are exposed as well. Finally, the current trends in this area are described.

1. INTRODUCTION

Linear digital signal/image processing techniques have been the primary tools used in digital signal/image processing and analysis. Their mathematical simplicity and the existence of a unifying linear systems theory made their design and implementation easy. Moreover, linear processing techniques offered satisfactory performance for a variety of applications. However, many digital signal/image processing and analysis problems cannot be solved by using linear techniques. Such examples will be given in subsequent paragraphs and are related to *nonlinearities due to noise and/or signal statistics*, to *system nonlinearities in digital signal/image acquisition, transmission and perception* and to *nonlinearities introduced by the objective goals and/or the processing method*. As a result of the shortcoming of linear techniques in all these cases, nonlinear digital signal/image processing and analysis emerged as early as 1958. Since then, it has had a dynamic development. This is indicated by the amount of research presently published and the popularity and widespread use of nonlinear digital processing in a variety of applications. Most of the currently available image processing software packages include nonlinear techniques (e.g. median filters, morphological filters). The purpose of this paper is to give an overview of the entire area of nonlinear digital signal/image processing and its development in the past two decades and to present the perspectives of this area in the years to come. Both goals are challenging, taking into account the impressive growth of this area and its fragmented and multi-faceted nature. In the following we give a non-exhaustive list of such system nonlinearities that lead to nonlinear processing.

1. Non-Gaussian noise/signal statistics. A primary example is impulsive noise filtering that frequently occurs in telecommunication channels, in TV broadcasting and in information losses in decoding or due to packet losses.

2. Signal-dependent noise. Multiplicative noise appears frequently during digital image acquisition. Speckle noise appears in ultrasonic images as well as in SAR images. Film grain noise can be modeled as multiplicative noise as well. Photoelectronic digital image acquisition (e.g. by using video cameras) introduces signal-dependent noise.

3. Non-stationarity in signal/image statistics. Digital image statistics change from one image region to the other. Adaptive (often nonlinear) operators are used to keep track of the varying image statistics.

4. Signal/image acquisition nonlinearities. Several acquisition devices are highly nonlinear systems (e.g. photographic film, video cameras), whose nonlinearities are of polynomial form. Furthermore, nonlinear (usually analog) preprocessing is used sometimes in order to increase the dynamic range of the acquisition, or for compensation reasons (e.g. *gamma* correction in television).

5. Transmission channel nonlinearities. Certain telecommunication channels introduce nonlinear (polynomial) distortion that produces nonlinear intersymbol interference and reduces the margin against noise. Quadratic and cubic nonlinearities provide a fairly good model for such channels.

6. Nonlinearities of the human perception mechanisms. In many occasions, the final judge of a digital signal/image processing system is human perception. It is widely known that perception mechanisms are very difficult to model. However, it has been found that certain of its parts have nonlinear models. For example, it is widely accepted that the human vision has logarithmic nonlinearities. Furthermore, high-level knowledge mechanisms that are decision-driven interfere with the low-level perception mechanisms. Decisions appear as hard nonlinearities (e.g. decision on edge detection) that lead to decision-directed nonlinear digital signal/image processing methods. In other cases, the characteristics of human visual perception lead to highly nonlinear digital image processing operations (e.g. histogram equalization, dithering).

7. Optimization of nonlinear objective measures. In several cases, nonlinear measures and/or features express fairly well certain signal/image characteristics. Image entropy is a primary example of a nonlinear measure of image information content (and tentatively its quality). Optimization of such measures leads to nonlinear digital signal processing (e.g. maximum entropy image restoration).

8. Nonlinearities in digital image analysis and in object recognition. Many digital image analysis operations use hard nonlinearities (e.g. maximum/minimum selection, thresholding, comparisons, quadratic nonlinearities), because image analysis leads to symbolic descriptions that are based on decisions. For example, decisions are needed for edge detection and for region split/merge. Furthermore, shape features (e.g. skeletons, convex hulls, linear piecewise approximations) are usually nonlinear functions of the respective digital images.

As can be seen in this list, nonlinear digital signal/image processing and analysis can tentatively cover a very broad and inhomogeneous area that is related, among others, to nonlinear systems theory, statistics, statistical communications theory, stochastic processes, pattern recognition and shape morphology. It is a melting pot of theories coming from all these areas as well as from classical digital signal/image processing applications (e.g. telecommunications, broadcasting, biomedical sciences, earth sciences, manufacturing and quality control). Thus, it has concrete theoretical and technological background. However, a source of criticism of this area is the lack of a unifying theory. This is a major theoretical and practical drawback in comparison

to linear digital signal processing and, at the same time, a great challenge ahead. There are historical reasons that explain the lack of an underlying theory and that are mainly related to the multitude of its theoretical background and to the fact that nonlinear systems theory is much more difficult to develop and generalize than linear systems theory. This does not mean that there were no efforts to formulate such unifying concepts. Some efforts have been useful in merging nonlinear filter classes (e.g. the theory of stack filters unified morphological and order statistics filter). Robust statistics tools have been successful in analyzing and merging classes of nonlinear filters (e.g. L-filters, R-filters and M-filters). However, a complete unifying theory is possibly far away from current efforts. This review paper will present classes of related nonlinear digital filters and to summarize their properties. The following classes of nonlinear digital filters will be presented:

- Order statistic filters
- Morphological filters
- Homomorphic filters
- Polynomial filters

because they have exhibited an impressive growth in the past two decades. Finally, an effort will be made to present recent promising trends in this area in the near future. It must be noted that the research area presented in this paper has a rich literature that cannot be presented here due to space limitations. The references included in this paper are only books and review papers.

2. NONLINEAR FILTERS BASED ON ORDER STATISTICS

The class of filters based on order statistics is very rich. The best known filter is the *median filter*. It originates from robust estimation theory. It was originally suggested for time series analysis. Later on, it became popular in digital image processing because of its computational simplicity and its good performance.

Let $X_1, X_2, \dots, X_n$ be random variables. If they are arranged in ascending order of magnitude $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ $X_{(i)}$ is called the i -th order statistic [1]-[8]. A very important order statistic is the *median* $med(X_i) = X_{(\nu+1)}$ (if $n = 2\nu + 1$). A *median filter* is a *running* median operator that scans the signal/image. The statistical and deterministic properties of the median filter have been studied extensively. It has been proven that median filters perform well for *long-tailed* noise distributions (e.g. Laplacian noise), whereas their performance is poor for *short-tailed* noise distributions (e.g. uniform noise) [2],[3]. The good performance of the median filter at long-tailed distributions is explained by the fact that the median is the maximum likelihood estimate (MLE) of location for the Laplacian distribution. The statistical and robustness properties of the median make it very suitable for impulsive noise filtering. Median filters can remove impulses with large probability of success.

Edge information is very important for human perception. Therefore, its preservation and, possibly, its enhancement is a very important subjective feature of the performance of an image filter. The median filter tends to preserve the edge sharpness, due to its robustness properties. The median filter not only smooths noise in homogeneous image regions, but tends to produce regions (called *streaks*) of constant or nearly constant intensity. These effects are undesirable because they are perceived as lines or contours, which do not exist in the original image. If streaking is a serious problem, the only solution is to use modifications of the median filter.

A very important tool in the analysis of linear systems is their steady state behavior with sinusoidal inputs. Of course, this is not valid for median filters, because they are highly nonlinear systems. Therefore, many researchers have focused on the problem of finding signals (called *roots* or *fixed points*), which are invariant under median filtering [2,4,5]. This problem and related ones constitute the *deterministic analysis* of median and related filters. This area has been very active in the past decade.

Since its first use, several modifications and extensions of the median filter have been proposed. Some of them (e.g. separable median filters, recursive median filters) are direct extensions. Other filters incorporate structural and time/spatial information in median filtering (e.g. weighted median filters, multistage median filters, median hybrid filters) [7]. Many extensions have a rather ad hoc structure, due to the lack of a powerful unifying theory in the area of nonlinear filtering. Several efforts have been made in the past decade in providing a unifying theory in the area of order statistics filtering. Some fruitful results based on *threshold decomposition* (to be described in this section) are expected to provide useful design and implementation tools.

It is attractive both from a practical and theoretical point of view to reduce median (and related) filters of k -valued signals to median (and related) filtering of binary signals, because binary filters can be used instead. This can be obtained by the *threshold decomposition* of such filters [8]. The M -valued signal x_i is decomposed by thresholding to $M - 1$ binary signals. These signals are processed independently by using binary filters (Boolean functions). The outputs $y_i^{(j)}, 1 \leq j \leq M - 1$ of these binary filters possess the so-called *stacking property* if the binary output signals are piled and the pile of the filter output y_i at time i consists of a column of 1's having a column of 0's on top. The filters that support the stacking property are called *stack filters*. Stack filters possess both the threshold decomposition structure and the stacking property. Stack filters are a broad class of nonlinear filters having median filters, filters based on order statistics and morphological filters as a special case. Great advances have been made recently in the design of stack filters [6,8]. An estimation approach and a structural approach have been proposed. The estimation approach employs the *Minimum Absolute Error (MAE)* criterion because of its robustness and its fitness to the thresholding decomposition.

The class of filters based on order statistics is very rich. Besides the median filter and its modifications, it includes a large number of nonlinear filters. Some of them (e.g. the ranked-order filters and the max/min filters) are straightforward applications of order statistics in filtering. Other filters (e.g. α -trimmed mean filters, L-filters) come from the application of robust estimation techniques in digital signal/image filtering. They are closely related to a large class of robust estimators called *L-estimators* [2],[9]. An important generalization of the median is the *L-filter*. The moving average, median, r -th ranked-order, α -trimmed mean and midpoint filters are special cases of the L-filter, if its coefficients are chosen appropriately. The filter coefficients can be chosen optimally according to the input noise probability function. As expected, the optimal L-filter for the Gaussian noise is the moving average filter. The optimal L-filter for the Laplacian distribution is close to the median. The optimal L-filter for the uniform distribution is the midpoint. L-filters can be extended to incorporate structural and time/spatial information leading to the so-called L1 filters (or C filters). Finally, order statistics filters are closely related to other classes of filters coming from robust estimation, notably R-filters (e.g. Wilcoxon filter) and M-filters [8,9]. It must be noted that robust estimation has provided very useful unifying tools in nonlinear digital filters (e.g. the notion of influence function and of the breakdown point).

One of the main reasons for the popularity and widespread use of some filters based on order statistics is their computational simplicity. Their computation can become faster, if appropriate fast algorithms are designed (notably *running* algorithms). Several such algorithms have ap-

peared in the literature, especially for the fast (serial or parallel) implementation of the median filter. Another research activity is the design of special VLSI chips for order statistics filtering. A number of chips for fast median and max/min filtering has been presented in the literature.

3. MATHEMATICAL MORPHOLOGY

Mathematical morphology has produced another important class of nonlinear digital signal/image processing and analysis operators [2], [10]-[17]. Its origin is shape analysis and its mathematical background is set theoretic. It is a powerful tool for geometrical shape analysis and description. It uses a set theoretic approach to image analysis. A binary image object X and its subparts are described by sets. Subpart interrelationships describe object structure. According to mathematical morphology principles, an image possesses no information if it is not observed by an observer. It is the observer, who examines the object and extracts the information needed for structure description and analysis. Observer-object interaction is obtained by using a simple set B (e.g. circle, square) that is called *structuring element*. The mode of interaction is a morphological transformation $\Psi(X)$. Its result can be a measure $m(\Psi(X))$ that describes aspects of the object structure. Information about the object size, shape, connectivity, smoothness and orientation can be obtained by using various structuring elements and morphological operators. The basic morphological transformations are *dilation* and *erosion*. It can be easily seen that these operators are very simple but yet very powerful. Their combination leads to more complex nonlinear operators like *opening* and *closing*. Morphological operations can be extended to functions and, therefore, to grayscale images [2,16,17]. The tools for grayscale morphological operations are simple functions $g(x)$ called *structuring functions*. If the structuring function $g(x)$ is zero, grayscale dilation and erosion are reduced to max, min filtering. Thus, grayscale morphological operations are closely related to filters based on order statistics that have been described in the previous sections. Morphological filters use binary or grayscale erosion and dilation as their basic building blocks. They are essentially special cases of the stack filters described in the previous section. They can be used efficiently in filtering impulsive noise. Some of them can also be used as edge detectors or peak detectors (e.g. top-hat transformation). However, the primary use of mathematical morphology techniques is on shape analysis [10]-[17]. It has been found that morphological operators can be efficiently used to thin objects and to find their *skeletons*. Morphological techniques have also been used to decompose grayscale images or binary shapes into simple components. A combination of mathematical morphology and Computer Solid Geometry (CSG) has been proposed for shape representation. Morphological methods (notably the so called *distance transforms*) have given elegant solutions to several problems of computational geometry (e.g. Voronoi tessellation, Delaunay triangulation, convex hull). Such methods can be considered to be the link between nonlinear digital signal/image processing, computational geometry and shape modeling. The list of the theoretical and technological achievements obtained by mathematical morphology is impressive, taken into account that it is based on extremely simple mathematical concepts.

4. HOMOMORPHIC FILTERS

In many applications, signals are convolved and/or are combined in a nonlinear way (e.g. in seismic signal processing, digital speech processing, signal transmission). In all these cases, classical linear techniques are not useful because the *superposition* property does not hold any more. Therefore, a special filter class has been developed for treating convolved and nonlinearly

related digital filters. They are called *homomorphic* filters [2,18]. They form one of the oldest nonlinear digital filter classes. Their basic goal is to transform two convolved or nonlinearly related signals $x_1(n) * x_2(n)$ to additive signals $\hat{x}_1(n)$, $\hat{x}_2(n)$ by using a nonlinear operator $D[x_1(n) * x_2(n)] = \hat{x}_1(n) + \hat{x}_2(n)$ and to process them by a linear filter $L[\hat{x}_1(n) + \hat{x}_2(n)] = \hat{y}_1(n) + \hat{y}_2(n)$. The output of the linear filter is transformed afterwards by the inverse nonlinearity $D^{-1}[\hat{y}_1(n) + \hat{y}_2(n)] = y_1(n) + y_2(n)$. Thus, the homomorphic systems satisfy the so called *generalized principle of superposition*. In the case of multiplicative nonlinearities of the form $x_1(n) * x_2(n) = x_1(n)x_2(n)$, the nonlinearity takes the form $D(x) = \ln(x)$ and the inverse nonlinearity is given by $D^{-1}(x) = \exp(x)$. The resulting classical logarithmic homomorphic system is used for multiplicative noise filtering of the form $x = sn$ that is frequently encountered in ultrasonic imaging, SAR imaging and can also be used to model cloud noise and illumination variations (s is the original image, n is the noise and x is the corrupted image). Wiener filters can be used in the place of the linear operator L . Homomorphic systems having nonlinearities $D(x) = x^\gamma$ can be used in signal dependent noise filtering of the form $x = c_1 s^l + c_2 s^m n$. Photoelectron noise encountered in digital image acquisition can be modeled by this form. Nonlinear mean filters can also be considered to be special cases of homomorphic systems, although their origin comes from statistics (harmonic mean, geometric mean, L_p mean) [2]. Homomorphic systems for convolution have a nonlinear operator D that is a cascade of a Z transform, a logarithm and an inverse Z transform. This transformation of the signal x produces its *cepstrum* $\hat{x}$. Complex cepstrum has extensively been used in speech and in geophysical signal processing, because convolved signals are extensively used in these applications.

5. POLYNOMIAL FILTERS

A classical way to represent the input-output relation of a nonlinear system (filter) that is amenable to characterization, analysis and synthesis is *Volterra* series representation. This can be considered as a Taylor series with memory. Although the class of nonlinear systems that can be represented by Volterra series is limited, they are extensively used because they lead to mathematically amenable representations. In many cases, the nonlinear systems are approximated by truncated Volterra series that result to relatively simple representations. The filters based on Volterra representation are frequently called *polynomial* filters. Their input-output relation is given by $y(n) = h_0 + \sum_{k=1}^{\infty} \sum_{i_1=0}^{N-1} \dots \sum_{i_k=0}^{N-1} h_k(i_1, \dots, i_k) x(n-i_1) \dots x(n-i_k)$ [19]. The term h_1 is the classical linear impulse response and the term $h_k(i_1, \dots, i_k)$ is the k -th order impulse response. In most cases the infinite sum is truncated to a maximal order K . If $K = 2$ the so called *quadratic* filters result [2,19,20]. Linear filters can be efficiently represented by linear transforms (Fourier, Laplace), due to their convolution property. Polynomial filters can also be represented by using multidimensional linear transforms, because their input-output relation can be expressed as a sum of multidimensional convolutions. Bispectra and higher-order spectra provide useful tools in analyzing the performance of polynomial filters.

Polynomial filters can be useful in many cases where linear filters are known to be suboptimal, such as in the equalization of nonlinear channels. They can also be applied for compensating typical nonlinearities that appear in the echo path and affect adaptive echo cancellers. They can also be used in modeling random signals that have (or are suspected to have) nonlinear behavior (e.g. as has been found in the case of daily stock returns). Linear systems modeling fails in such cases. Quadratic filters can be used in a variety of detection problems (e.g. in the optimal detection of Gaussian signal in Gaussian noise). Finally, polynomial filters have recently been proposed for digital image enhancement, edge extraction and nonlinear image/image sequence prediction.

6. CURRENT TRENDS IN NONLINEAR FILTERING

It is always difficult to present the current trends of a broad research area in brief. It is even more difficult to predict its evolution in the decade to come. However, this paper will try to identify such trends that are either theory-driven or technology-driven. Such trends are the following:

Multichannel signal/image processing. Nonlinear multichannel digital filters are no straightforward extensions of their single-channel counterparts. For example the definition of multivariate data ordering possesses interesting problems in the design of order statistics filters. Extensions of multichannel Volterra filters could be of interest as well.

Image sequence processing. The spatiotemporal nature of the image sequences poses new theoretical and computational problems in the design of image sequence filters. Furthermore, the human perception of image sequences puts new subjective qualitative goals in their digital treatment.

Adaptive nonlinear filters. Adaptive versions have been proposed for a variety of nonlinear digital filters [8,21]. However, the problems related to filter adaptation are far from been solved due to the nature of the adaptation problem (mainly in digital image processing), to the nature of nonlinear filter parameters (e.g. window size and shape) and to problems in adaptation convergence.

Neural filters. Neural networks are a major research trend that has influenced digital signal processing. It is believed that its merging with nonlinear filter classes will lead to useful filters that will have complex "learning" capabilities. For example, Volterra nonlinearities have already been used in neural networks [22].

Development of unifying theories. It is rather optimistic to state that a major unifying theory will appear soon. However, smaller steps aiming at integrating nonlinear filter families may prove very fruitful, as has already happened in the past.

As a conclusion, we can state that the area of nonlinear digital signal/image processing will be very active in the future. Our optimism is based on the fact that linear filters have been extensively studied and they are reaching a maturity that diverts many researchers to new areas. Nonlinear digital signal/image processing is such a challenging area. Furthermore, it is now well known that linear systems have poor performance in many cases and that more complex (possibly nonlinear) solutions are highly desirable, despite the fact that their performance is not understood completely. Thus, the growth of nonlinear digital signal/image processing is both theory-driven and application-driven. Therefore, there is a strong motivation in confronting the great challenges ahead.

References

- [1] H.A. David, *Order Statistics*, Wiley, 1981.
- [2] I. Pitas, A.N. Venetsanopoulos, *Nonlinear digital filters: principles and applications*, Kluwer Academic, 1990.
- [3] B.I. Justusson, "Median filtering: statistical properties" in *Two-dimensional digital signal processing II*, T.S. Huang editor, Springer Verlag, 1981.
- [4] G.R. Arce, N.C. Gallagher, T.A. Nodes, "Median filters: theory for one- and two-dimensional filters", in *Advances in Computer Vision and Image Processing*, T.S. Huang editor, JAI Press, 1986.

- [5] S.G Tyan, "Median filtering: Deterministic properties", *Two- dimensional Signal Processing II* , T.S. Huang editor, Springer Verlag, 1981.
- [6] E.J. Coyle, J.H.Lin, M. Gabbouj, "Optimal stack filtering and the estimation and structural approaches to image processing", *IEEE Trans. on Acoustics, Speech and Signal Processing*, vol.37, no.12, pp. 2037-2066, Dec. 1989.
- [7] T. Viero, Y. Neuvo, "3-D median structures for image sequence filtering and coding", in *Motion analysis and image sequence processing*, M.I. Sezan, R.I. Lagendijk editors, Kluwer Academic, 1993.
- [8] I. Pitas, A.N. Venetsanopoulos, "Order statistics in digital image processing", *IEEE Proceedings*, Dec. 1992.
- [9] P. S. Huber, *Robust statistics*, J.Wiley, 1981.
- [10] J. Serra, *Image Analysis and Mathematical Morphology* , Academic 1982.
- [11] J. Serra (editor), *Image analysis and mathematical morphology: theoretical advances*, vol. 2, Academic Press, 1988.
- [12] C.R. Giardina, E.R. Dougherty, *Morphological methods in image and signal processing*, Prentice Hall, 1988.
- [13] E.R. Dougherty, C.R. Giardina, *Image processing - continuous to discrete*, vol. 1, Prentice-Hall, 1987.
- [14] J. Serra, "Introduction to mathematical morphology", *Computer Vision, Graphics and Image Processing*, vol. 35, pp. 283-305, 1986.
- [15] R.M. Haralick, S.R. Sternberg, X. Zhuang, "Image analysis using mathematical morphology", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-9, no. 4, pp. 532-550, July 1987.
- [16] P. Maragos, "Tutorial on advances in morphological image processing and analysis", *Optical Engineering*, vol. 26, pp. 623-632, July 1987.
- [17] P. Maragos, R.W. Schafer, "Morphological systems for multidimensional signal processing", *Proceedings IEEE*, vol. 78, No.4, pp. 690-710, April 1990.
- [18] A.V. Oppenheim, R.W. Schafer, *Discrete-time signal processing*, Prentice-Hall, 1989.
- [19] M. Schetzen, *The Volterra and Wiener Theories of Nonlinear Systems* , Wiley 1980.
- [20] G.L. Sicuranza, "Quadratic filters for signal processing", *Proceedings of IEEE*, vol.80, pp.1263-1285, Aug.1992.
- [21] V.J. Mathews, "Adaptive polynomial filters", *IEEE Signal Processing Magazine*, pp.10-26, July 1991.
- [22] N.B. Karayannis, A.N. Venetsanopoulos, *Artificial neural networks: learning algorithms, performance evaluation and applications*, Kluwer Academic, 1992.